

Fungal Endophyte Community Analysis — LOT2 (Peru)

Analysis environment: R 4.6.1 | Generated on 2026-09-01

Overview

This project analyses the fungal endophyte communities isolated from three substrates collected in **Peru** (LOT2 sampling campaign):

Substrate	Isolates (pooled)	Sample-level replicates (units)
Ficus leaves	264	10
Ficus wood	167	12
Lauraceae leaves	219	8

Sampling design

Samples were collected at different **tree zones** (heights):

- **Zones 1-5:** Tree trunk (zone 1 = lowest, zone 5 = highest)
- **Zone 6:** Canopy branches (highest zone)

Wood collected from zone 6 corresponds to **branch wood** (no trunk present). Wood from zones 1-5 is **trunk wood**.

Substrate	Zones sampled	Notes
Ficus leaves	6 only	All leaf samples from canopy
Ficus wood	1-6	Trunk (zones 1-5) and branch (zone 6)
Lauraceae leaves	5-6	Predominantly zone 6

Handling of uncertain taxonomy

Entries with NA, "?", "N0", or "incertae sedis" are pooled into “**Incertae sedis**” before aggregation.

Input Data

- **LOT2_pooled_counts.xlsx** (first sheet) — Pooled genotype counts per substrate with full taxonomy (284 genotypes)
- **LOT2_samples.xlsx** (first sheet) — Individual isolate records with sampling zone and unit (650 isolates)

Scripts

Script	Purpose
main.R	Master script — loads libraries, sources all other scripts in order
parse_LOT2.R	Reads both Excel files (first sheet only), standardises column names
plot_abundance.R	Horizontal bar charts of isolate counts by taxonomic level
plot_abundance_pooled.R	Bar charts with uncertain taxa pooled as “Incertae sedis”
plot_pie.R	Pie charts of community composition (3 pies per level)
community_analysis.R	Full community ecology analyses across all comparisons
generate_readme.R	Generates this README dynamically from analysis results

```
# From the project directory:  
source("main.R")
```

R packages used

Package	Version	Role
readxl	1.5.0	Reading Excel input
dplyr	1.2.1	Data manipulation
tidyr	1.3.2	Data reshaping
ggplot2	4.0.3	Plotting
scales	1.4.0	Axis formatting
vegan	2.7.5	Diversity, NMDS, PERMANOVA, ANOSIM, betadisper
indicspecies	1.8.0	Indicator species analysis (IndVal)
ggVennDiagram	1.5.7	Venn diagrams

Output Structure

```
plots/  
  abundance/                # Bar charts (raw)  
  abundance_pooled_incertae_sedis/ # Bar charts (uncertain taxa pooled)  
  pie_charts/               # Pie charts  
  community/                # Community ecology plots (NMDS, rarefaction, etc.)  
  png/                      # PNG versions for README embedding
```

Results

A. Alpha Diversity (from pooled counts)

Index	What it measures	Higher means
Richness (S)	Number of distinct taxa	More taxa present
Shannon (H')	Combines richness and evenness	More diverse
Simpson (1-D)	Probability two random individuals differ	More diverse
Inverse Simpson	Effective number of equally-common species	More even
Pielou (J')	How evenly individuals are distributed	More even

At ITS taxon level

Substrate	S	N	H'	1-D	Inv. Simp.	J'
Lauraceae leaves	106	219	4.2745	0.9779	45.289	0.9166
Ficus leaves	135	264	4.4705	0.9791	47.8681	0.9114
Ficus wood	81	167	4.0073	0.9722	35.9858	0.9119

Full data: `tables/alpha_diversity.csv`

B. Community Composition

B1. Rank-Abundance Curves

B2. Relative Abundance

B3. Venn Diagrams — Shared Taxa

At **genus level** (44 total genera): **7** shared across all 3 substrates, **2** unique to Lauraceae leaves, **10** unique to Ficus leaves, **20** unique to Ficus wood.

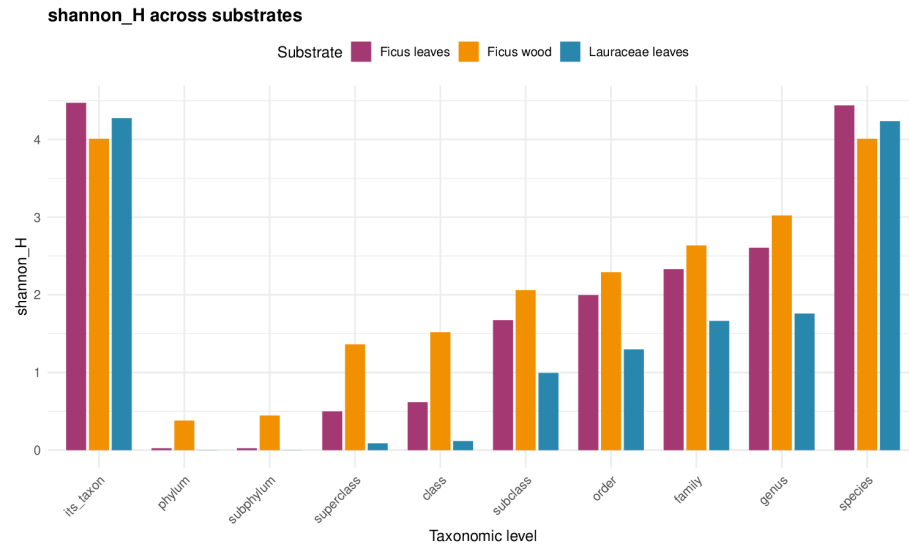


Figure 1: Shannon diversity

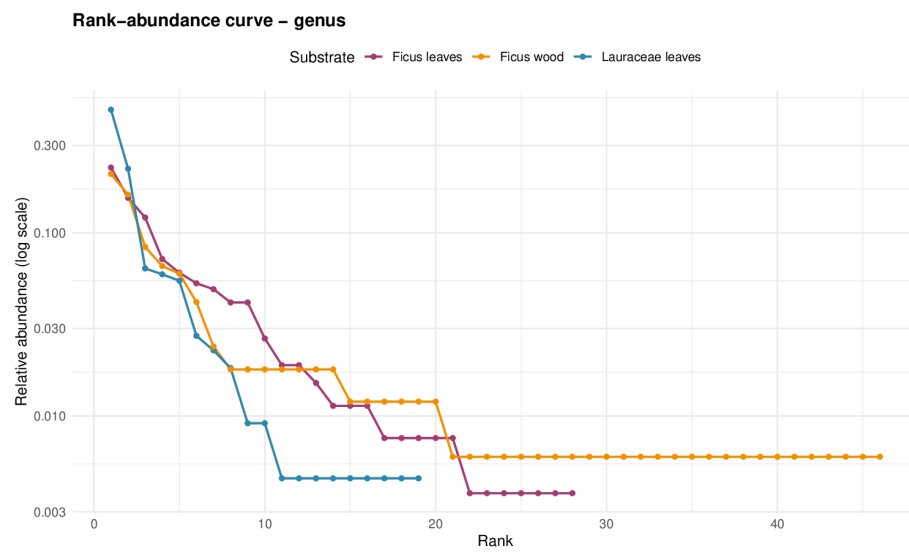


Figure 2: Rank-abundance at genus level

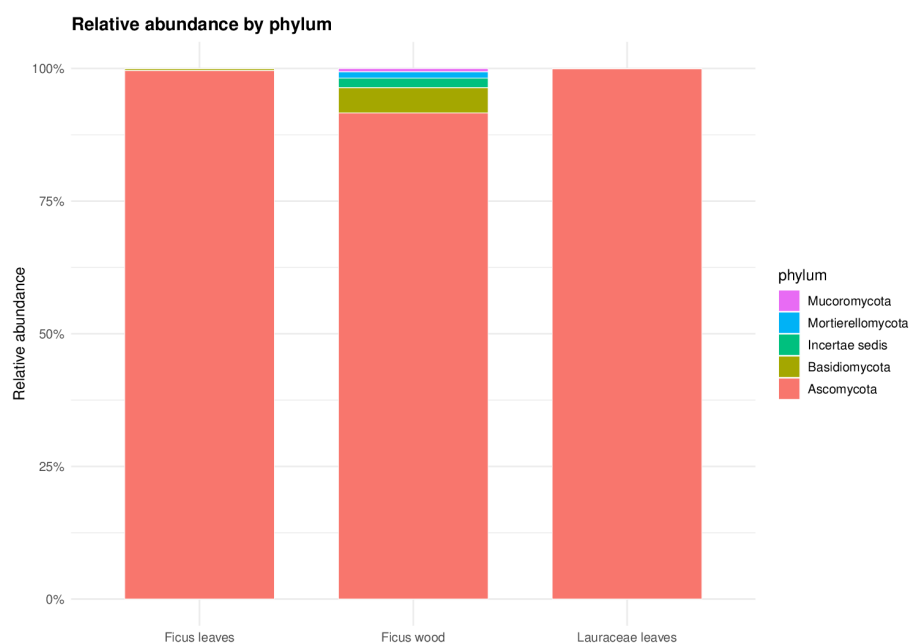


Figure 3: Relative abundance by phylum

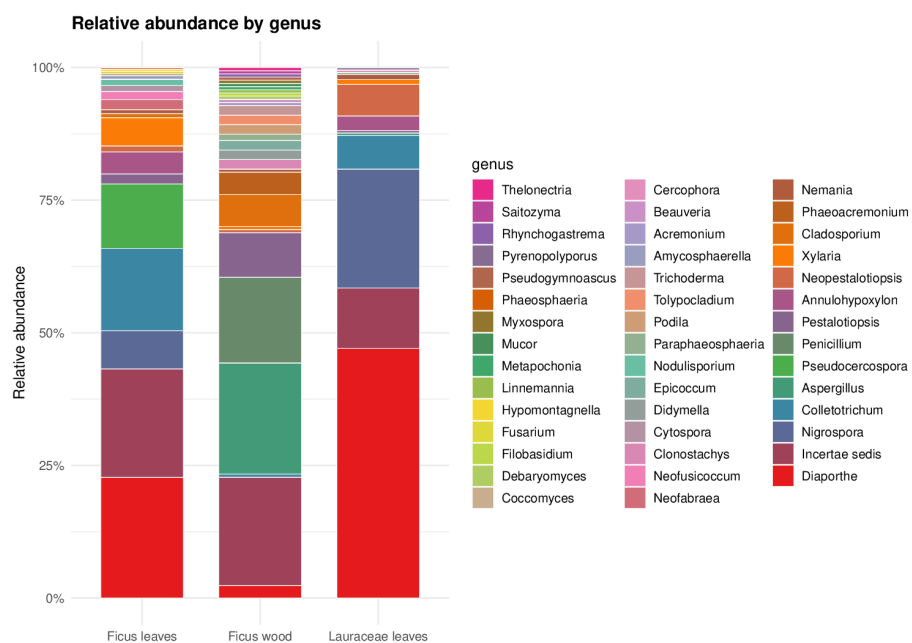


Figure 4: Relative abundance by genus

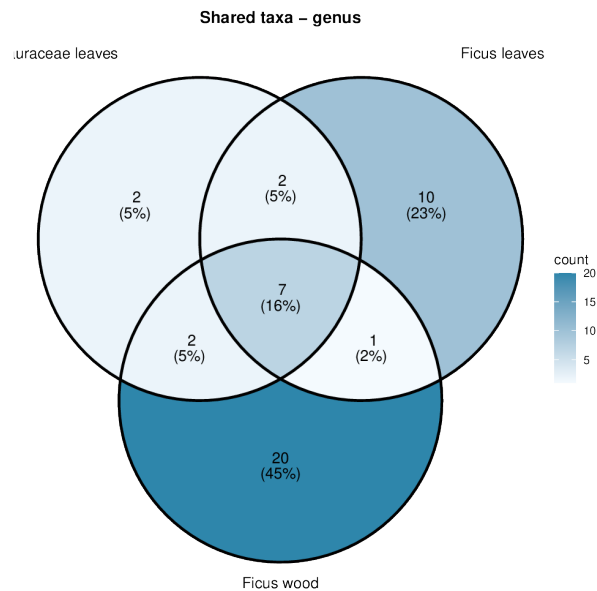


Figure 5: Venn diagram — genus

Multivariate Statistical Analyses

Sample-level analyses use **32 samples** and **284 ITS taxa**, with **Bray-Curtis dissimilarity** and **999 permutations**.

C1. Substrate Comparison: Ficus Leaves vs Ficus Wood vs Lauraceae Leaves

NMDS Ordination

Stress = **0.0771** (< 0.10 = good; < 0.20 = acceptable)

PERMANOVA

PERMANOVA - Bray-Curtis distance

Formula: community ~ substrate

Permutations: 999

Permutation test for adonis under reduced model

Permutation: free

Number of permutations: 999

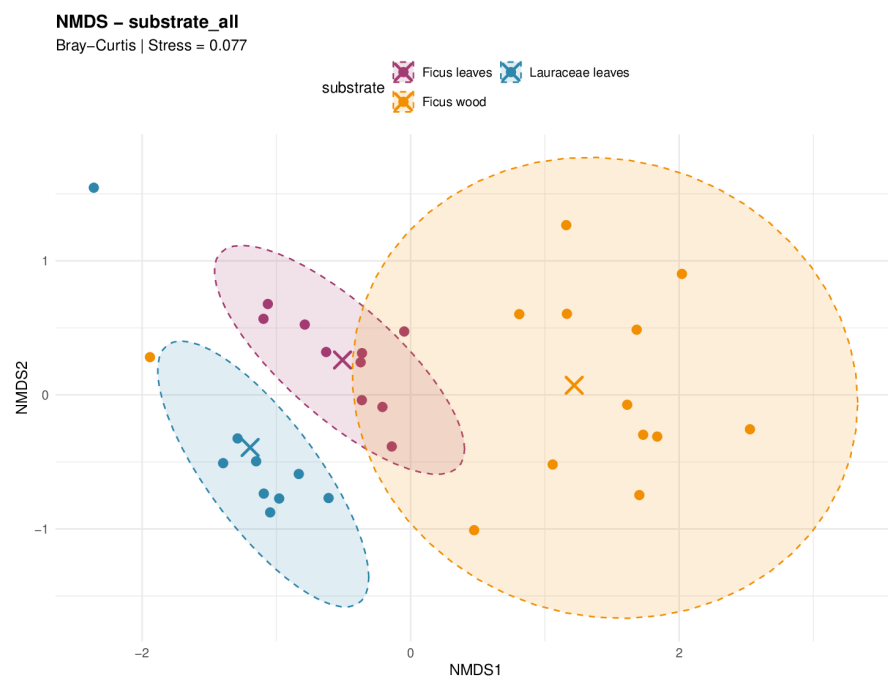


Figure 6: NMDS — all substrates

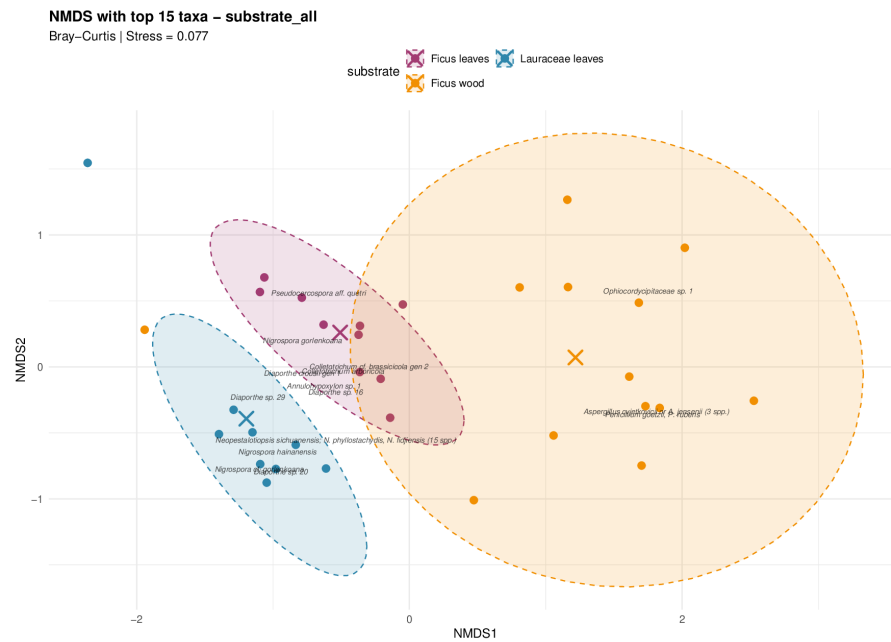


Figure 7: NMDS with species overlay

```
adonis2(formula = form, data = meta, permutations = 999, method = "bray")
      Df SumOfSqs      R2      F Pr(>F)
Model    2    2.792 0.20041 3.6342  0.001 ***
Residual 29   11.140 0.79959
Total    31   13.932 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F = 3.6342, R2 = 0.20041, p = 0.001
```

ANOSIM

R = 0.5604 , p = 0.001

Beta-dispersion (PERMDISP)

F = 4.332, p = 0.02

Pairwise PERMANOVA

Comparison	F	R ²	p (raw)	p (Holm)	Sig.
Ficus leaves vs Ficus wood	3.8123	0.1536	0.001	0.003	**
Ficus leaves vs Lauraceae leaves	3.3294	0.1638	0.001	0.003	**
Ficus wood vs Lauraceae leaves	3.6839	0.1555	0.001	0.003	**

Rarefaction

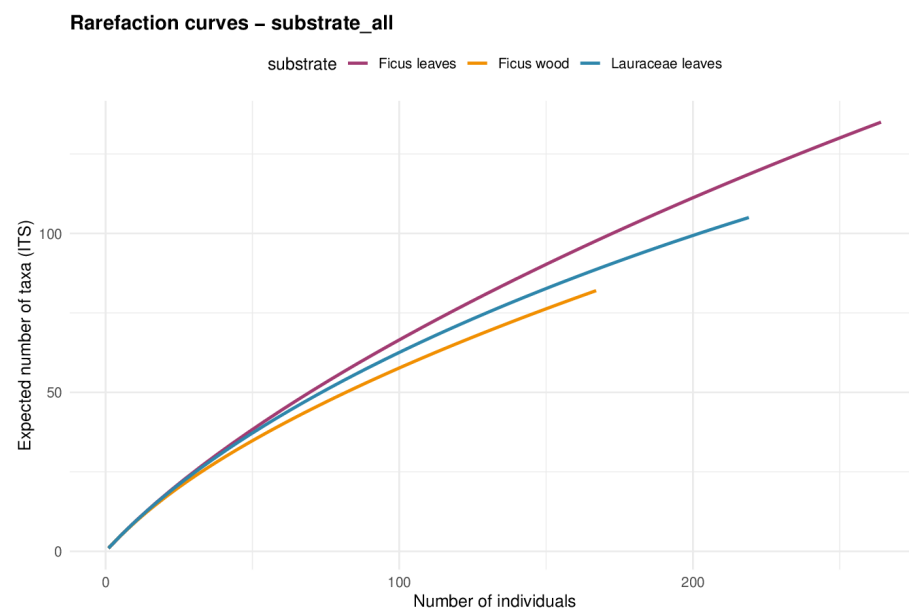


Figure 8: Rarefaction — all substrates

Indicator Species (23 significant, $p = 0.05$)

Full results: `tables/indicator_species_significant.csv`

C2. Ficus Leaves vs Lauraceae Leaves

PERMANOVA: $F = 3.3294$, $R^2 = 0.16377$, $p = 0.001$

Full results in `tables/substrate_leaves_*.txt`

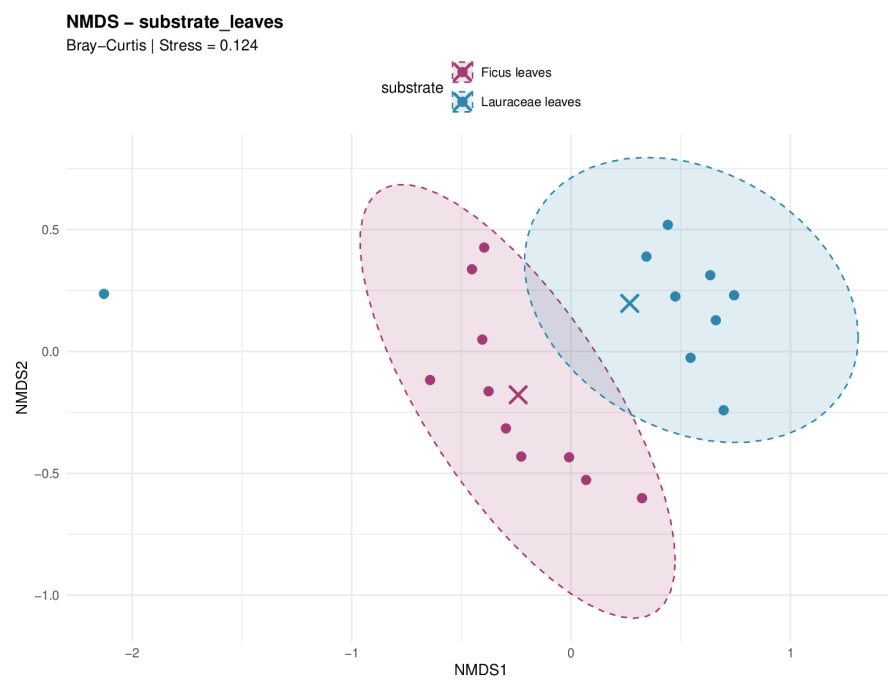


Figure 9: NMDS — Ficus vs Lauraceae leaves

E. Zone-Based Analyses

E1. Ficus Wood Across Zones

Community comparison of Ficus wood isolates collected at different tree zones (1-6).

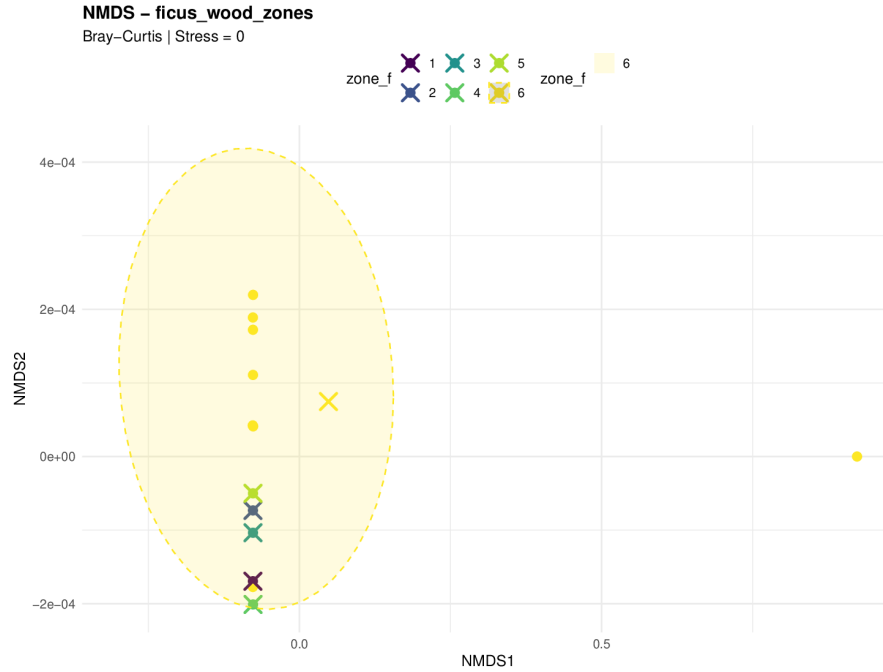


Figure 10: NMDS — Ficus wood zones

Full results in `tables/ficus_wood_zones_*.txt`

E2. Ficus Wood: Trunk (Zones 1-5) vs Branch (Zone 6)

PERMANOVA: $F = 1.9802$, $R^2 = 0.15256$, $p = 0.004$

E4. Substrate Comparison at Zone 6 (Branch Level Only)

PERMANOVA: $F = 3.3279$, $R^2 = 0.22443$, $p = 0.001$

E5. Substrate $\times$ Position Interaction

Combined factor analysis testing whether community composition differs by substrate-position combinations.

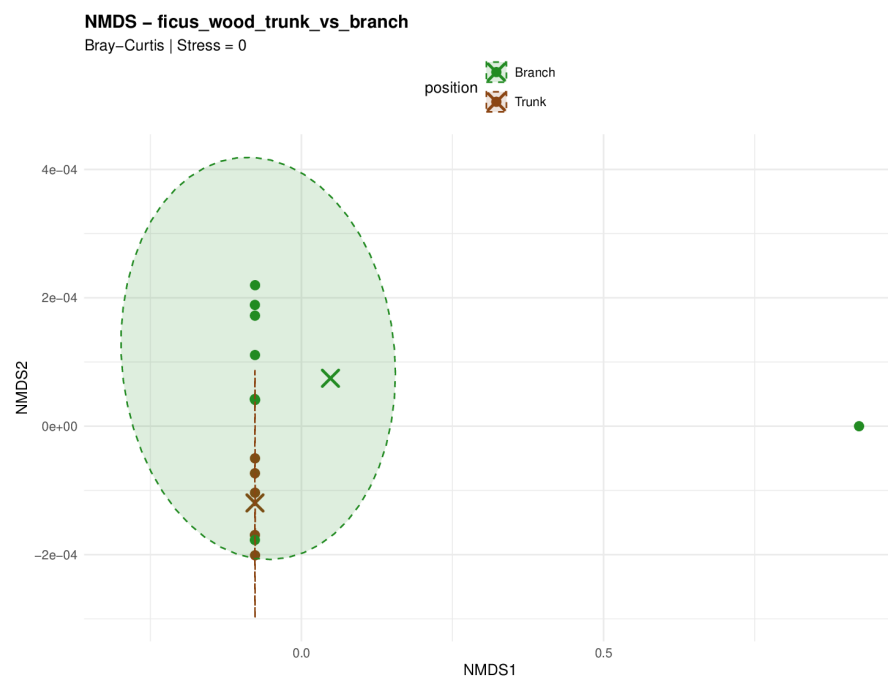


Figure 11: NMDS — trunk vs branch

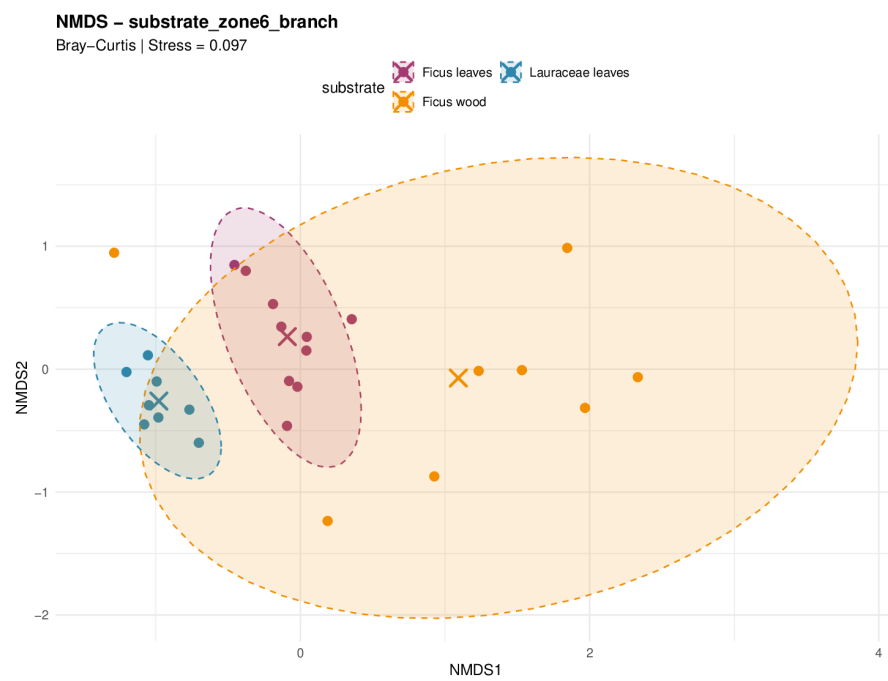


Figure 12: NMDS — substrates at zone 6

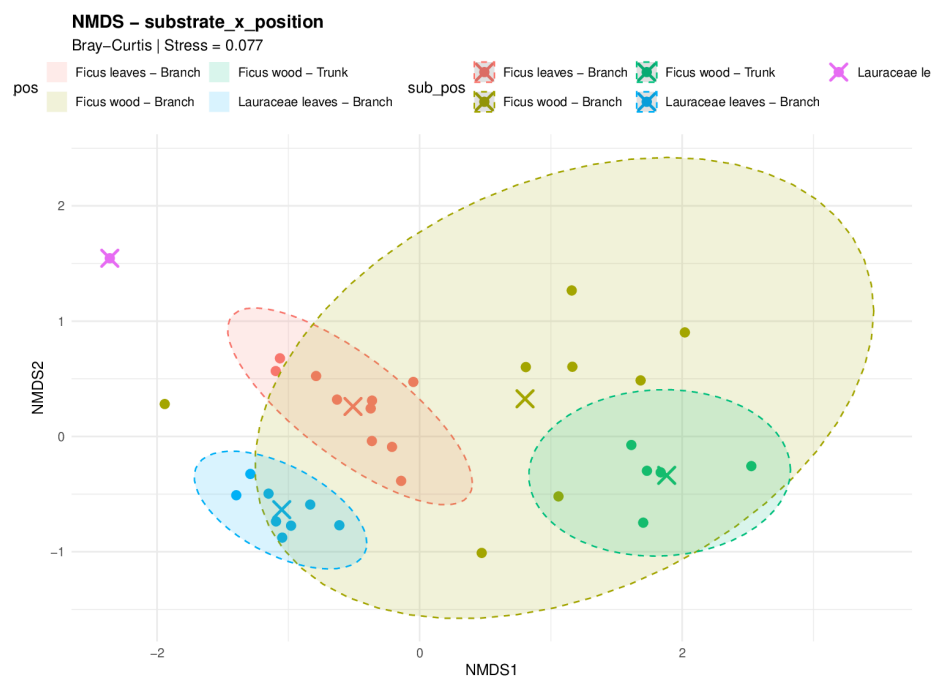


Figure 13: NMDS — substrate × position

Abundance Plots

plots/abundance/

Horizontal grouped bar charts showing absolute isolate counts per taxonomic level, broken down by substrate.

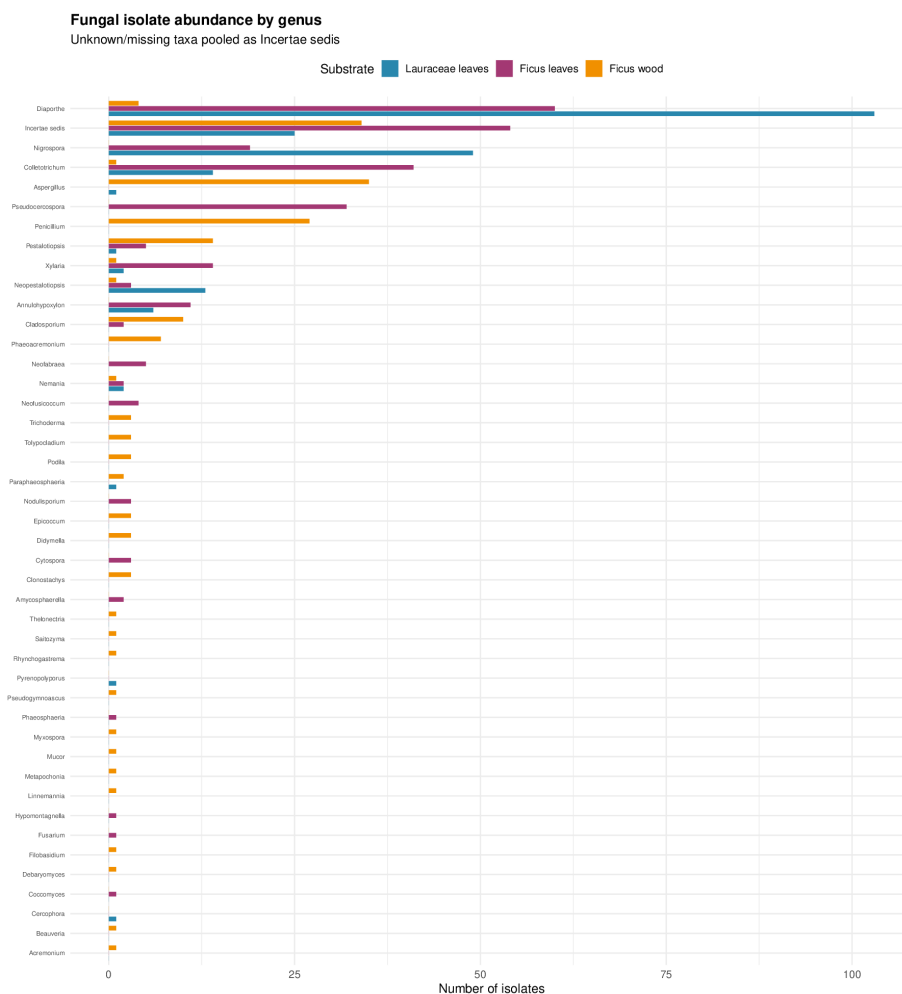


Figure 14: Abundance by genus (pooled)

plots/pie_charts/

Auto-generated on 2026-09-01 by generate_readme.R

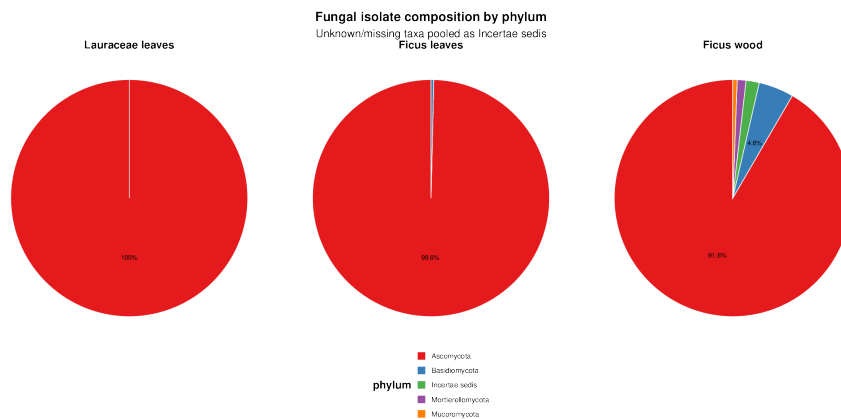


Figure 15: Pie chart by phylum

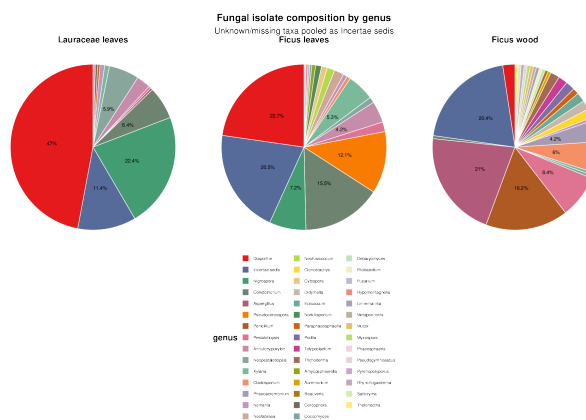


Figure 16: Pie chart by genus